

ABSTRACT

The ShieldPatch platform provides a comprehensive, easy-to-use, secure way to automatically detect vulnerabilities in software and files, and manage the patching process. The primary goal of ShieldPatch is to assist both individuals and organizations in rapidly identifying vulnerabilities and minimizing their risk of being exploited. To accomplish this, ShieldPatch utilizes both machine learning and live threat intelligence feeds to analyze and evaluate the risk of a given vulnerability, and recommend or deploy appropriate patches. A real-time dashboard provides users with an effective method for monitoring their systems status, evaluating vulnerability analysis results, and obtaining suitable patching solutions for their systems. The purpose of ShieldPatch is to reduce the time elapsed between the discovery of a vulnerability and the availability of a suitable patch to remediate that vulnerability, thereby improving security and enabling safer computing environments through reduced manual intervention.